

University of the South Pacific

School of Information Technology, Engineering, Mathematics, and Physics

IS328 - Data Mining

Short Test Sample Solution

Semester 2, 2025

Possible Marks:	30
Weight:	5%
Time Allowed:	50 minutes
Number of Pages:	6 (including this page)

Learning Outcomes:

1. Explain the concepts, definitions and processes of Data Mining

Instructions:

1. Answer ALL questions.
2. Write ALL the answers in the spaces provided in the Answer Booklet.
3. If you require extra space for your answers, use the last page.

Q1: Pre-Processing (20 marks)

a) What is the difference between Data Mining and Query Processing in databases?

[2 marks]

- Query Processing: Retrieves **explicitly stored information** from a database.
- Data Mining: Discovers **hidden patterns, trends, or knowledge not directly** stored in the database.

b) The table below shows sample attributes from a university database. For each attribute, identify its data type and provide a short justification [5 marks]

Student_ID	GPA_Category	Scholarship_Awarded	Tuition_Fees_FJD	Admission_Year
S101	High	Yes	2500.50	2018
S102	Medium	No	3000.00	2020
S103	Low	Yes	3200.75	2023

Attribute	Data Type	Justification
Student_ID	Nominal	Acts as a unique identifier/label only; no inherent order or measurable value.
GPA_Category	Ordinal	Ordered categories (High > Medium > Low), but the difference between them isn't measurable.
Scholarship_Awarded	Asymmetric Binary	Two possible values (Yes/No). "Yes" is meaningful, while "No" is absence, so it is asymmetric.
Tuition_Fees_FJD	Ratio	Continuous numeric; has a true zero (0 FJD = no fee), and ratios/differences are meaningful.
Admission_Year	Interval	Ordered numeric values where differences matter (2020–2018 = 2 years), but ratios are meaningless (2020 ≠ 2 × 1010).

- c) During data preprocessing, why is data cleaning considered one of the most important steps in the Knowledge Discovery in Databases (KDD) process? Provide two common issues that data cleaning resolves. [3 marks]

Importance: Data cleaning ensures the dataset is accurate, consistent, and complete before analysis, which improves the quality of patterns discovered.

Issues resolved:

1. Missing values (e.g., no age recorded for a customer).
2. Duplicate or inconsistent entries (e.g., same customer listed twice with slightly different names).

- d) A bank is preparing its customer dataset for churn analysis. The following preprocessing actions are taken. For each action, identify the data preprocessing task performed. [4 marks]

Action	Description	Preprocessing Task
1	Converting all customer balances from USD to FJD	Data Transformation
2	Selecting only the top 10 most relevant features	Data Reduction (Feature Selection)
3	Grouping ages into categories: Youth, Adult, Senior	Data Discretization
4	Combining online banking data with branch transaction data	Data Integration

- e) Given the dataset: [2, 4, 6, 8, 10, 12, 14, 16]; perform equal-depth (equal-frequency) binning with 4 bins and apply smoothing by bin mean. Briefly explain why smoothing is applied in data preprocessing. [3 marks]

Equal-depth binning (2 values per bin):

- Bin 1: [2, 4] → mean = 3 → [3, 3]
- Bin 2: [6, 8] → mean = 7 → [7, 7]
- Bin 3: [10, 12] → mean = 11 → [11, 11]
- Bin 4: [14, 16] → mean = 15 → [15, 15]

Final Smoothed Data: [3, 3, 7, 7, 11, 11, 15, 15]

Why smoothing is applied:

- To reduce noise and fluctuations, creating clearer, more stable patterns for mining.

f) Given the dataset: [5, 10, 15], normalize it using Z-score normalization, assuming population standard deviation. [5 marks]

- Mean: $\mu = (5+10+15)/3 = 30/3 = 10$
- Population $\sigma = \sqrt{[(5-10)^2 + (10-10)^2 + (15-10)^2] \div 3}$
 $= \sqrt{[(25+0+25)/3]} = \sqrt{(50/3)} \approx 4.08$
- Z-scores:
 - $z(5) = (5-10)/4.08 \approx -1.22$
 - $z(10) = (10-10)/4.08 = 0$
 - $z(15) = (15-10)/4.08 \approx +1.22$
- Normalized Data: [-1.22, 0, +1.22]

Q2: Classification Techniques (10 marks)

a) What is the main difference between the ZeroR and OneR classification algorithm?

[2 marks]

- ZeroR: Predicts the majority class only and does not use any attributes.
- OneR: Builds a classification rule using a single attribute (the one that yields the lowest error).

b) The dataset below shows students' Internet Access and Exam Result.

Example	Internet Access	Exam Result
S1	WiFi	Pass
S2	WiFi	Fail
S3	WiFi	Pass
S4	Mobile	Fail
S5	Mobile	Fail
S6	None	Fail
S7	None	Pass

i. For each value of *Internet Access*, identify the frequent class by constructing a frequency table. [1 mark]

Internet Access	Pass	Fail	Frequent class
WiFi	2	1	Pass
Mobile	0	2	Fail
None	1	1	Fail (tie → choose Fail)

ii. Calculate the total number of classification errors if OneR uses this attribute.

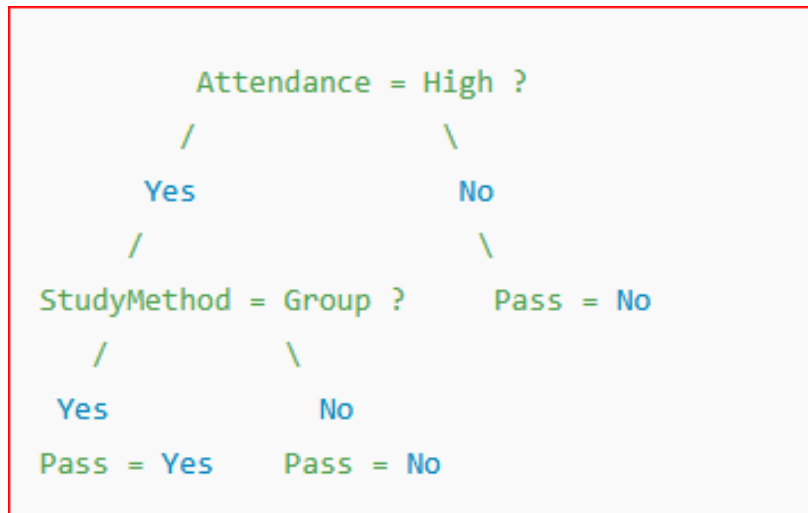
[1 mark]

- WiFi: predict **Pass** → errors = 1 (the single Fail)
 - Mobile: predict **Fail** → errors = 0
 - None: predict **Fail** → errors = 1 (the single Pass)
- Total errors = 1 + 0 + 1 = 2**

iii. State the final OneR rule. [1 mark]

If Internet = WiFi → Pass
If Internet = Mobile → Fail
If Internet = None → Fail

c) Based on the decision tree below, generate all possible IF-THEN rules that can be used to predict the outcomes. [2 marks]



1. IF Attendance = High AND StudyMethod = Group → Pass = Yes
2. IF Attendance = High AND StudyMethod ≠ Group → Pass = No
3. IF Attendance = Low → Pass = No
4. (Implicit default rule) IF Attendance is missing → Pass = No

d) Decision trees sometimes suffer from overfitting. What does overfitting mean in the context of decision trees? [1 mark]

Overfitting means the tree becomes too complex and fits noise in the training data rather than capturing general patterns.

e) What is a confusion matrix in classification, and why is it useful? [2 marks]

A confusion matrix is an n*n table that compares **actual vs. predicted classes** of a classifier.

It is useful because it helps evaluate performance beyond accuracy, showing details like **true positives, false positives, true negatives, and false negatives**.